

NITHISH REDDY DEVULAPALLI

DevOps Engineer

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CAREER OBJECTIVE

An IT professional with over 4.5 years of hands-on experience in **DevOps**, specializing in **AWS** and **Azure** cloud environments, my expertise lies in configuring and optimizing various cloud services, including **EC2, S3, RDS, Lambda, Virtual Machines, App Services, and Storage**. Proficient in infrastructure provisioning through **AWS CloudFormation** and **Azure Resource Manager** templates, I leverage **Docker** and **Kubernetes** for efficient application deployment. My strong suit includes the design of robust **CI/CD** pipelines, utilizing tools like **Jenkins, GitLab CI/CD, and Azure DevOps**, coupled with automation using **PowerShell, Bash, and Python**. Furthermore, I bring proficiency in **configuration management tools, version control systems, and Linux/Unix administration**, all underpinned by a commitment to effective communication and collaboration to deliver high-quality solutions.

SKILLS

CLOUD ENVIRONMENTS: AWS, Azure, GCP.

INFRASTRUCTURE AS A CODE (IaC) TOOLS: Terraform, Cloud Formation, ARM Templates, GCP Deployment Manager.

CI/CD TOOLS: Jenkins, Azure DevOps, GitHub Actions, GitLab, VSTS.

CONTAINERIZATION TOOLS: Docker, Kubernetes (EKS, GKE, AKS), Helm, KOPS, Kubectl

CONFIGURATION MANAGEMENT TOOLS: Ansible, Chef, Puppet.

MONITORING TOOLS: CloudWatch, CloudTrail, Prometheus, Grafana, ELK Stack, OpenSearch, Kafka, Splunk, App Insights.

VERSION CONTROL TOOLS: Git, GitHub, Bitbucket, Azure Repos.

SCRIPTING LANGUAGES: Python, Bash, PowerShell, Perl, Java, YAML, Groovy.

EXPERIENCE

SR. CLOUD/DEVOPS ENGINEER

Wells Fargo | Cary, NC | May 2024 – Present

- Strong experience in **Infrastructure Development** and **Operations** involving **AWS Cloud Platforms, EC2, IAM, EBS, ECS, S3, Glacier, VPC, Red Shift, RDS, SES, ELB, Auto Scaling, CloudWatch, Cloud Front, Cloud Formation, Elastic Cache, SNS, SQS, Lambda, ROUTE 53, and AWS** relational databases like **Aurora, MYSQL, MSSQL and NoSQL Database DynamoDB**.
- Developed an **IaC** framework for cloud migration, automating workflows for support teams, and architecting **CI/CD** infrastructure with **Jenkins** and **Bitbucket Webhooks**. Created Various automation Scripts using **Python** to automate daily routine jobs like changing the instance types, uploading and downloading the objects without using the AWS console.
- Configured **AWS CloudWatch** with **DataDog** for monitoring AWS Resources with Second to Second and created custom Dashboards and metrics by using **Python**. Enabled **Detailed CloudWatch** Monitoring for **RDS Oracle Database** and stored metrics about at **1-minute** granularity and created alarms based on metrics.
- Creating **Roles** for AWS Services and Groups for users using **Identity Access Management** and enabling **MFA** for **Users**. Implemented Jenkins pipelines for infrastructure automation, created Python modules for API integrations, and documented CI/CD workflows in Confluence to enhance team efficiency.

- Creating **S3** buckets and restricting access to buckets and directories to specific **IAM users**. Used **ELBs**, to load balance the front-end requests to the app servers, and to load balance the backend requests from the app servers to the **API servers**.
- Worked in writing Templates for **AWS** infrastructure as a code using **Terraform** and **Cloud Formation** to build staging and production environments. Defined **Terraform** modules such as **Compute, Network, Operations**, and **Users** to reuse in different environments.
- Worked on POC for Deploying the **AWS** infrastructure using **Terraform, CloudFormation**. Created and configured Redshift Cluster for data-warehousing responsible for Security, included opening different ports on security groups and **Network ACL**.
- Performed the automation deployments using **AWS** by creating **IAM** roles and used the code pipeline plug-in to integrate **Jenkins** with **AWS** to create the **EC2** instances to provision the virtual Servers. Extensively worked on Jenkins for continuous integration and for End-to-End automation for all build and deployments. Set up Continuous Integration for major releases in Jenkins.
- Designed and configured **Azure App - Cloud Services, PaaS, Azure Data Factory, Azure Blob Storage, Web API, VM creation, ARM Templates, PowerShell scripts, IAAS, storage, network**, and **database**.
- Configured **Azure** Virtual Networks, subnets, DHCP address blocks, Azure network settings, DNS settings, security policies and routing. Also, deployed **Azure IaaS** virtual machines and **Cloud services** (PaaS role instances) into secure Virtual Networks and subnet.
- Set up Jenkins server and built jobs to provide continuous automated builds based on polling the **GIT SCM** during the day and periodically scheduled the builds overnight to support development.
- Created **Jenkins** jobs for continuously building the projects. Scheduled jobs to build upon every check in into subversion. Wrote Groovy script in **Jenkins** jobs to do pre-and post-build operations.
- Worked with Docker on multiple cloud providers like **AWS, Azure, GCP** from helping developers build and containerize their applications (**CI/CD**) to deploying either on public or private cloud.
- Implemented container-based deployments using **Docker** and designed Docker images, Hubs and registries and installed Docker Registry for local upload, download of Docker images and Docker hub.
- Configured cluster operations in **Amazon Elastic Kubernetes Service (EKS)** to deploy microservices along with **Kubernetes** with **CI/CD** system and used **Kubernetes** cluster components to scale up operations of clusters, maintain the cluster services, ingress, network policies and to group Docker containers across different platform.
- Worked on **Docker** with **Kubernetes** to create pods for applications and implemented **Kubernetes** to deploy a web application across a Multi-node **Kubernetes** cluster.
- Used **GIT** as a source code management tool for creating local repo, cloning the repo, adding, committing, pushing the changes in the local repo, saving changes for later (Stash), recovering files, branching, creating tags.
- Worked on **GIT**, Bitbucket to track changes made by different people in source code and configured **GIT** with **Jenkins** and schedule jobs using **POLL SCM** option.
- Used **Drone** to fetch the projects from **GIT** and written drone.yml files to set up and run the internal applications.

AWS DEVOPS ENGINEER

DXC Technology | Hyderabad, India | Mar 2020 – Dec 2022

- Experience in managing and maintaining **IAM Policies** for organizations in **AWS** to define groups, create **Users**, Assign Roles, and define rules for **Role-Based Access to AWS Resources**.
- Experience on **AWS**, focusing on **High-Availability, Fault Tolerance**, and **Auto-Scaling** using **Terraform Templates**. Along with (**CI/CD**) with **AWS Lambda** and **AWS Code Pipeline** and using **Python** included **Boto3** to supplement automation provided by **Ansible and Terraform** for tasks such as encrypting **EBS Volumes** backing **AMIs** and scheduling **Lambda** functions for routine **AWS** tasks.
- Utilize **Terraform** expertise to develop and version infrastructure on **Azure** and **AWS** with confidence. Utilized Azure/AWS Terraform modules to create resources, automate infrastructure management, and gain experience with key **Terraform** features like **Infrastructure as Code, Execution Plans, Resource Graphs, Change Automation**, and **Auto Scaling** for launching cloud instances while deploying **Microservices**. I also used **Terraform** to map more complex dependencies and locate network issues.
- Expertise on using **Terraform** for automating **VPCs, ELBs, Security Groups, SQS Queues, S3** buckets, and continuing to replace the rest of our infrastructure and managed different infrastructure cloud resources, **VMware**, Bare Metal Servers, and **Docker Containers**.

- Experience in using Kubernetes for creating new projects, services for **Load Balancing**, and adding them to **Routes** by accessing from outside, created **Pods** through new application and controlling, scaling, and troubleshooting pods through SSH, managed manifest files and releases of **Helm** charts.
- Experience with container-based deployments using **Docker**, worked on **Docker Files**, **Docker Images**, **Docker Hub**, **Docker Registries** and **Kubernetes**.
- Used **Kubernetes** to manage **Containerized** applications utilizing its **Nodes**, **Config-Maps**, and **Services**, and application containers were deployed as **Pods**.
- Designing and implementing **CI (Continuous Integration)** system: configuring **Jenkins Servers**, **Jenkins Nodes**, **TeamCity**, creating required scripts (**Perl & Python**), and creating/configuring servers (**Windows/Linux**).
- Integrated and streamed **AWS CloudWatch** Logs to **Splunk** by triggering **AWS Lambda** and pushing events to **Splunk** for real-time analysis and visualization.
- Hands-on experience with **Ansible** scripts and **Ansible Tower** as **Configuration** management tool, to automate repetitive tasks, quickly deploy critical applications and cloud infrastructure.
- Performed **Cloning**, modifying, updating the **GIT** repos and push back to the **Bitbucket**. Imported and managed multiple corporate applications into **GIT** repositories and give access rights to the authorized developers.
- Experience with **Administration/Maintenance** of **Source Control Management Systems** such as **GIT**, **Subversion (SVN)** on **Linux** and windows platform. Knowledge on setting up project baselines, **Branching**, **Merging**, and **Tagging** source code in **GIT**.
- Administered **Linux** servers for several functions including managing **Apache Tomcat** server, mail servers, and **Oracle**, **MongoDB**, **MySQL** databases in both development and production.
- Creation of **Python** backend and **JavaScript** frontend components integrating with the firm's enterprise management systems.

SITE RELIABILITY ENGINEER

Wipro Limited | Bengaluru, India | Jan 2019 – Mar 2020

- Extensively worked with various **Azure Services** like **Web Roles**, **Worker Roles**, **Azure Websites**, **Caching**, **Net Worker Services**, **API Management**, **Active Directory (AD) Services** infrastructure in advocating, maintaining, and monitoring. **Azure Active Directory (AAD)** infrastructure incorporated with periodic auditing, troubleshooting and performance.
- Experience in designing, configuring, and deploying solutions on **Microsoft Azure** using (Azure Resource Manager) **ARM Templates**, **Azure PowerShell Module** and, **Azure CLI** focusing on high-availability and auto-scaling.
- Proficient in **Microsoft Azure IAAS**, **SAAS** and **PAAS** services such as **Virtual Networks**, **Virtual Machines**, **Blob Storage**, **Azure Data Factory**, **Cloud Services**, **Resource Groups**, **Express Route**, **Traffic Manager**, **VPN**, **Load Balancing**, **Application Gateways**, **Auto-Scaling**, deploying **Web Apps**, monitoring by **Azure Application Insights**, and creating **Web-Jobs**.
- Worked on container management using **Docker** by writing **Docker Files** and set up the automated build on **Docker Hub**. Written **Docker Compose** file for multi container provisioning and make file to build, run, tag, and publish a docker container to **Azure Container Registry**.
- Created cloud modules for interacting with **Azure** services which provides the tools to easily create and orchestrate infrastructure on **Azure** and automated cloud-native applications in **Azure** using **Azure Microservices** such as **Azure Functions** and **Kubernetes** on **Azure Ansible**.
- Managed **Azure Infrastructure Azure Web Roles**, **Worker Roles**, **SQL Azure**, **Azure Storage**, **Azure AD Licenses**, **Office365**. **Virtual Machine Backup** and **Recover** from a **Recovery Services** Vault using **Azure PowerShell**.
- Experience in configuring **Azure Traffic Manager**, analyzing **Azure Networks** with **Azure Network Watcher**, **Implementing Azure Web Apps**, **Azure App Services**, **Azure Application Insights**, **Azure Application Gateway**, **Azure DNS**, **Azure Stack**, **Azure Backup** and **Azure Automation**.
- Expertise in deploying **Micro Services**, its application improvement and upgradation utilizing **Azure** administrations like **Azure DevOps**, **Kubernetes Service (AKS)**, **Container Services**, **MySQL DB**, and **Grafana**, **Azure Pipelines**, **RBAC**, and **Checking Groups**.
- Implemented a **CI/CD** pipeline using **Azure DevOps (VSTS, TFS)** in both cloud and on-premises with **GIT**, **MS Build**, **Docker**, **Maven** along with **Jenkins Plugins**.
- Implemented **Azure DevOps Pipelines** to drive all micro services builds out to the **Docker** registry and then deployed to **Kubernetes**, **Created Pods** and managed using **AKS**.
- Created **Network Security Groups (NSG)** to filter the traffic to and from **Azure Resources** by configuring the **Inbound** and **Outbound** rules and associating them with **Subnet** and **Network Interfaces**.

- Deployed **Kubernetes** cluster on **Azure** with master architecture and wrote many **YAML** files to create many services like **Pods, Deployments, Auto-Scaling, Load Balancers, Labels, Health Checks, Namespaces, Config Map**, etc.
- Used **Terraform** for building, changing, versioning infrastructure and collaborated the automation of **Azure Infrastructure** via **Terraform** and **Jenkins**.
- Implemented **Jenkins** pipeline jobs to create **Azure Infrastructure** from **GitHub** repositories containing **Terraform Code** and created on-premises active directory authentication using automation with ansible play books.
- Worked on creating, maintaining, and troubleshooting **LVM** and file management. Administration of repetitive jobs using **Cron** scheduling and using **Bash** scripts to accomplish the tasks on multiple servers.

EDUCATION

Stevens Institute of Technology

Master's in Management and Information Systems

Hoboken, NJ, US, December 2024

RELEVANT COURSEWORK

Database Management, Data Analytics & Machine Learning, Engineering Programming: Python, Financial Lab:

Database Design, Introduction to R, Intro to Python for Financial App